

# Track Name

Track: Game Development / Interactive Systems

Category: 2D Arcade Game using Python (Pygame)

## Setup & Run Instructions

1. Install Python (3.10 or higher) Check: `python --version`

2. Download the folder from GitHub and unzip it.

Link: <https://github.com/AbidMd/Python/tree/main/Game>

3. Navigate to project folder `cd Game`

4. Create Virtual Environment `python -m venv venv`

5. Activate Virtual Environment Windows: `venv\Scripts\activate`

6. Install Requirements `pip install -r requirements.txt`

7. Run the Game

`python main.py`

## Feature List

- Snake movement in the snake is controlled using keyboard arrow keys, allowing smooth directional movement within a grid-based environment. The movement system ensures responsive control and prevents illegal reverse direction inputs
- Food spawning items appear at random positions on the grid at regular intervals. Each time the snake consumes food, it grows longer, increasing the difficulty of navigation.
- Score tracking the game maintains a real-time score that increases as the snake eats food. This provides feedback to the player and encourages higher performance with each session.
- The game over detection continuously checks for failure conditions such as collisions. Once triggered, the game stops and the display score go back to zero.

## Screenshots

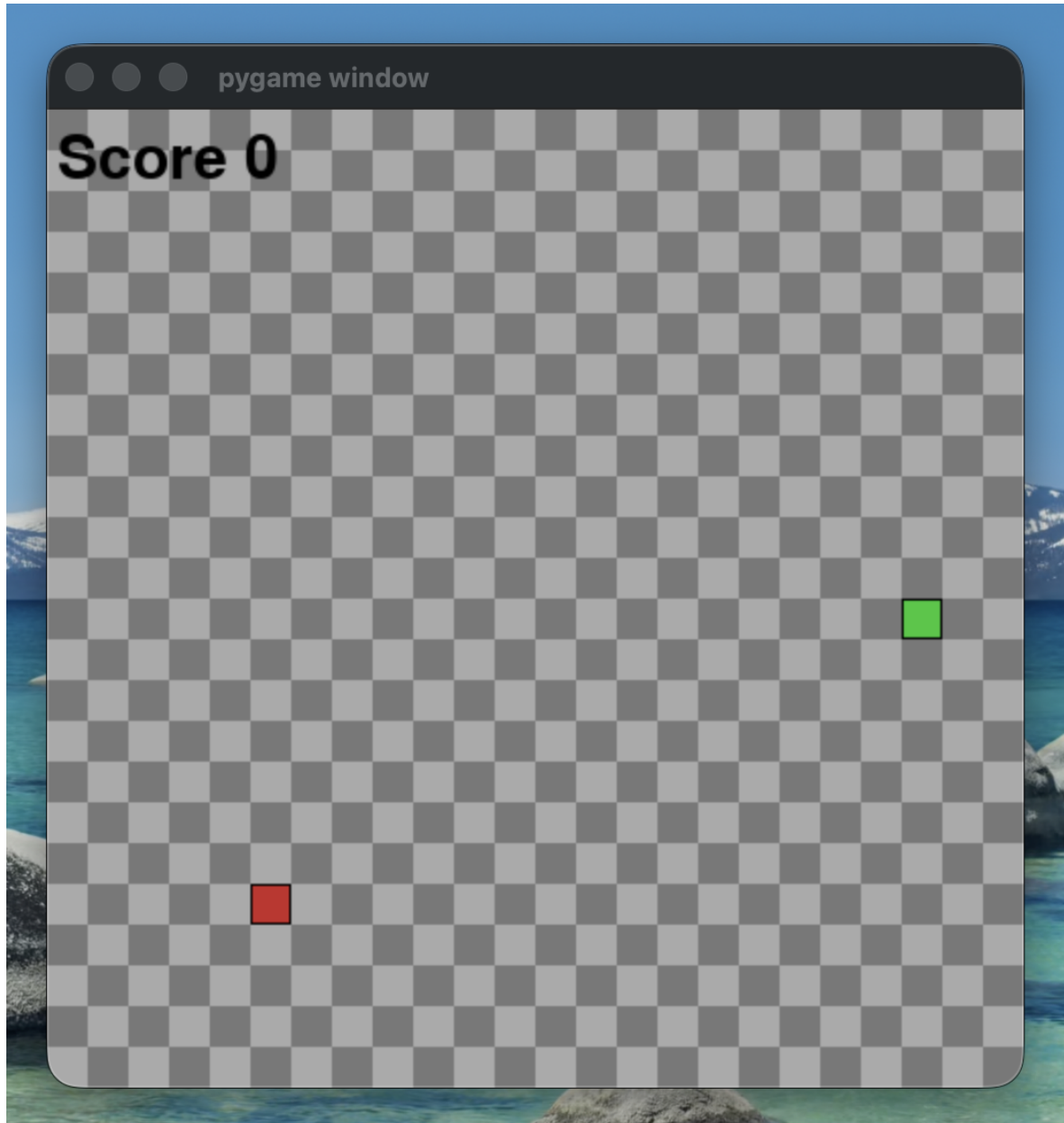


Figure 1: Game starts from score Zero

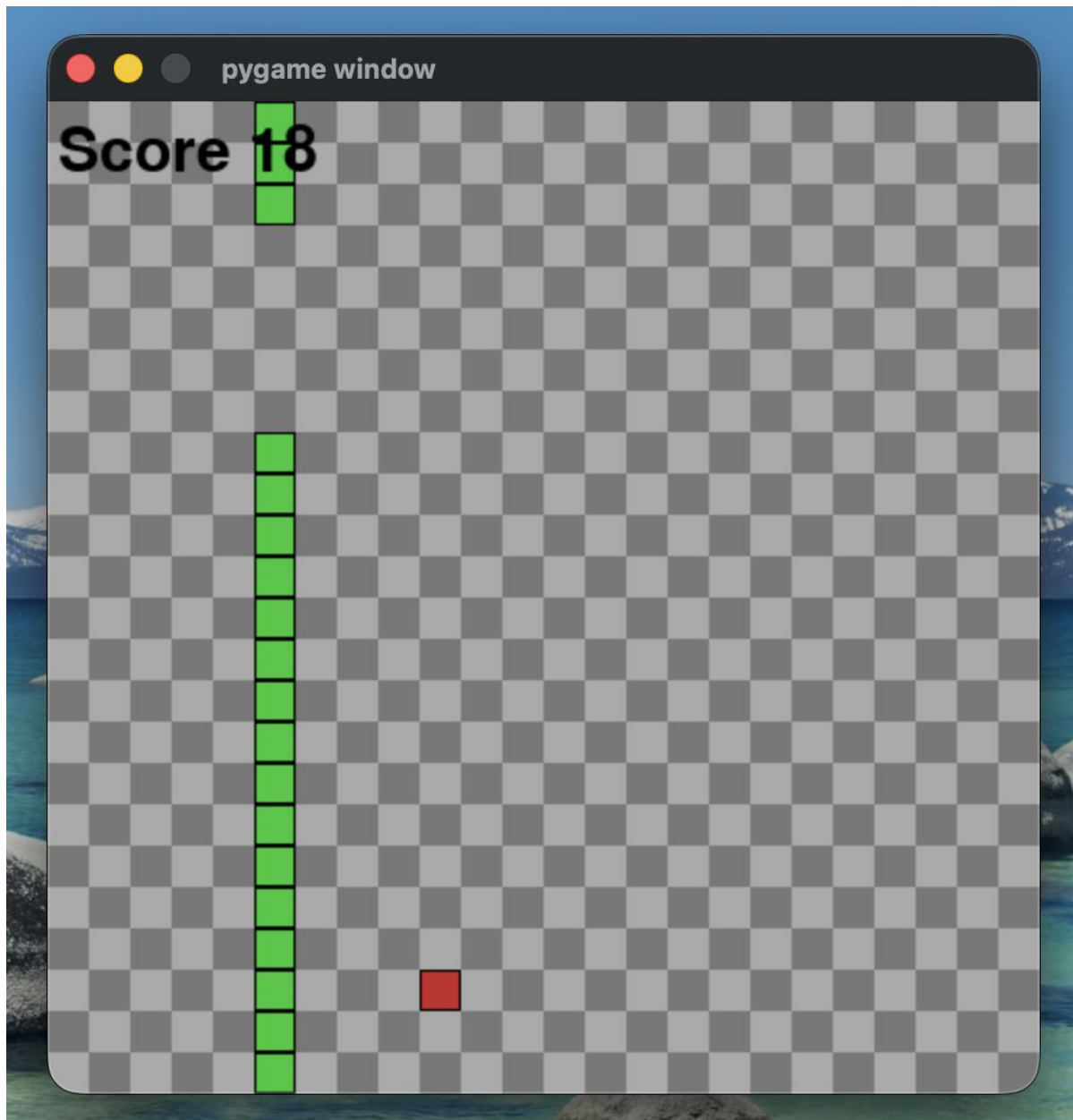


Figure 2: Players can enjoy it if the snake does not hit itself. When they hit itself, the score goes back to zero again.